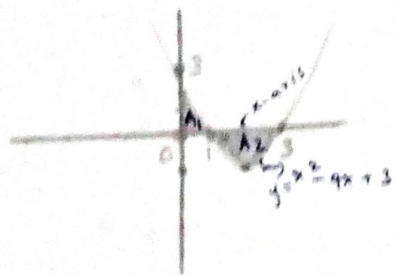


1) Find the area of the region between the x -axis and the graph of $f(x) = x^2 - 4x + 3$ over $[0, 3]$.



$$y=0 \quad y = x^2 - 4x + 3$$

$$x^2 - 4x + 3 = 0$$

$$(x-1)(x-3) = 0$$

$$x=1, x=3$$

$$\text{Area} = \int_a^b f(x) - g(x) dx$$

$$A_1 = \int_1^3 (x^2 - 4x + 3) dx$$

$$= \left[\frac{x^3}{3} - 4\frac{x^2}{2} + 3x \right]_1^3$$

$$= \left(\frac{27}{3} - 2(9) + 3(3) \right) - (0)$$

$$= \frac{1}{3} + 1 = \frac{4}{3} \text{ sq. units}$$

$$A_2 = \int_0^1 0 - (x^2 - 4x + 3) dx$$

$$= - \left[\frac{x^3}{3} - 2x^2 + 3x \right]_0^1$$

$$= - \left(\frac{1^3}{3} - 2(1^2) + 3(1) \right) - (0)$$

$$= - \left(\frac{1}{3} - 2 + 3 \right)$$

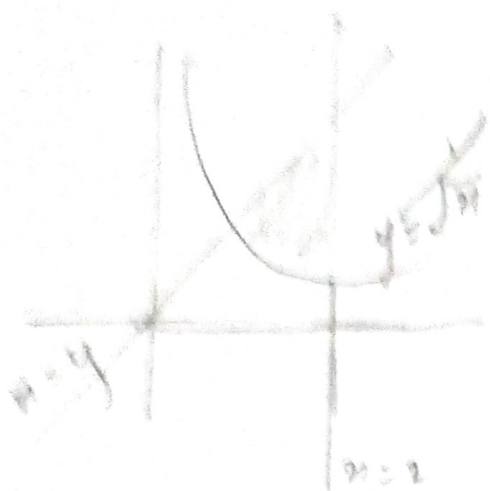
$$= - \left(\frac{1}{3} \right) = \frac{1}{3} \text{ sq. units}$$

$$\text{Total Area} = A_1 + A_2$$

$$= \frac{4}{3} + \frac{1}{3}$$

$$= \frac{5}{3} \text{ sq. units}$$

2) Find the area of the region bounded by the curves
 $y = x$, $y = \frac{1}{\sqrt{x}}$ and $x = 2$



$$y = x \quad y = \frac{1}{\sqrt{x}}$$

$$x = \frac{1}{\sqrt{x}}$$

$$x\sqrt{x} = 1$$

$$x^{3/2} = 1$$

$$x = 1$$

$$a = 1 \quad b = 2$$

$$\text{Area} = \int_a^b f(x) - g(x) dx$$

$$= \int_1^2 x - \frac{1}{\sqrt{x}} dx$$

$$= \int_1^2 x - x^{-1/2} dx$$

$$= \left[\frac{x^2}{2} - \frac{x^{-1/2+1}}{-1/2+1} \right]_1^2$$

$$= \left(\frac{2^2}{2} - \frac{2^{1/2}}{1/2} \right) - \left(\frac{1^2}{2} - \frac{1^{1/2}}{1/2} \right)$$

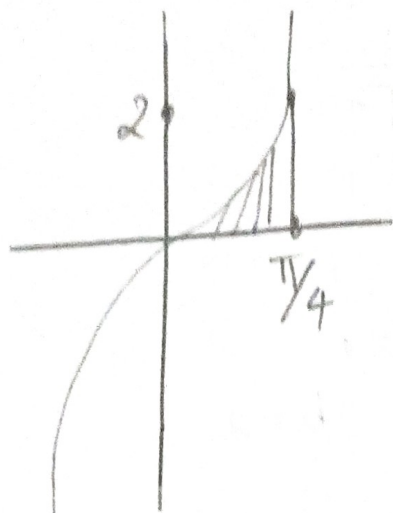
$$= 2 - 2\sqrt{2} - \frac{1}{2} + 2$$

$$= \frac{7}{2} - 2\sqrt{2}$$

$$= \frac{7 - 4\sqrt{2}}{2} \text{ sq units}$$

$$\text{Area} = \left(\frac{7 - 4\sqrt{2}}{2} \right) \text{ sq units}$$

2) Find the volume of the solid generated by revolving the region bounded by the curves $y = 2 \tan x$, $y = 0$ & $x = \frac{\pi}{4}$ around the line $x = \frac{\pi}{4}$.

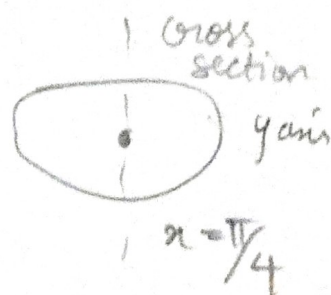


$$y = 2 \tan x \quad x = \frac{\pi}{4}$$

$$y = 2 \tan \frac{\pi}{4}$$

$$y = 2$$

$$\left(\frac{\pi}{4}, 2 \right)$$



$$2 \tan x = y \Rightarrow x = \tan^{-1} \left(\frac{y}{2} \right)$$

$$\text{radius } r = \frac{\pi}{4} - \tan^{-1} \left(\frac{y}{2} \right)$$

$$V = \int_0^2 \pi \left(\frac{\pi}{4} - \tan^{-1} \left(\frac{y}{2} \right) \right)^2 dy$$

4. Find the arc length of $y = \left(\frac{5}{12}\right)x^{6/5} - \left(\frac{5}{8}\right)x^{4/5}$ from $x=1$ to $x=32$

$$L = \int_a^b \sqrt{1 + (f'(x))^2} dx$$

$$a=1 \quad b=32$$

$$y = \frac{5}{12} x^{6/5} - \frac{5}{8} x^{4/5}$$

$$y' = \frac{5}{12} \cdot \frac{6}{5} x^{6/5-1} - \frac{5}{8} \cdot \frac{4}{5} x^{4/5-1}$$

$$= \frac{1}{2} x^{1/5} - \frac{1}{2} x^{-1/5} = \frac{1}{2} (x^{1/5} - x^{-1/5})$$

$$(f'(x))^2 = \frac{1}{4} (x^{1/5} - x^{-1/5})^2$$

$$= \frac{1}{4} (x^{2/5} - 2x^{1/5} x^{-1/5} + x^{-2/5})$$

$$= \frac{x^{2/5}}{4} - \frac{1}{2} + \frac{x^{-2/5}}{4}$$

$$1 + (f'(x))^2 = 1 + \frac{x^{2/5}}{4} - \frac{1}{2} + \frac{x^{-2/5}}{4}$$

$$= \frac{x^{2/5}}{4} + \frac{1}{2} + \frac{x^{-2/5}}{4} = \frac{1}{4} (x^{2/5} + 2 + x^{-2/5})$$

$$= \frac{1}{4} \left(x^{2/5} + 2 + x^{-2/5} \right)$$

$$= \frac{1}{4} \left(x^{2/5} + 2x^{1/5} \cdot x^{-1/5} + x^{-2/5} \right)$$

$$= \left(\frac{1}{2} \left(x^{1/5} + x^{-1/5} \right) \right)^2$$

$$\sqrt{1 + (f'(x))^2} = \sqrt{\left(\frac{1}{2} \left(x^{1/5} + x^{-1/5} \right) \right)^2}$$

$$= \frac{1}{2} \left(x^{1/5} + x^{-1/5} \right)$$

$$L = \int_a^b \sqrt{1 + (f'(x))^2} \, dx$$

$$= \int_1^{32} \frac{1}{2} \left(x^{1/5} + x^{-1/5} \right) \, dx$$

$$= \frac{1}{2} \left[\frac{5x^{6/5}}{6} + \frac{5x^{4/5}}{4} \right]_1^{32}$$

$$= \frac{5}{4} \left[\frac{x^{6/5}}{3} + \frac{x^{4/5}}{2} \right]_1^{32}$$

$$= \frac{5}{4} \left(\frac{(32)^{6/5}}{3} + \frac{(32)^{4/5}}{2} - \left(\frac{1}{3} + \frac{1}{2} \right) \right)$$

$$= \frac{5}{4} \left(\frac{2(5\sqrt{32})^6}{6} + \frac{3(5\sqrt{32})^4}{6} - \left(\frac{2+3}{6} \right) \right)$$

$$= \frac{5}{4} \left(\frac{2(2^6) + 3(2^4)}{6} - \frac{5}{6} \right)$$

$$= \frac{5}{4} \left(\frac{64 \times 2 + 16 \times 3 - 5}{6} \right)$$

$$= \frac{5}{4} \left(\frac{128 + 48 - 5}{6} \right) = \frac{5}{4} \left(\frac{171}{6} \right)$$

$$= \frac{5}{4} \left(\frac{57}{2} \right) = \frac{285}{8} = 35.625 \text{ units}$$

length of arc = 35.625 units

5. Find the area of the surface revolving the curve $x = \sqrt{4y - y^2}$, $1 \leq y \leq 2$ about the y-axis.

$$S = \int_a^b 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy$$

$$a=1 \quad b=2 \quad x = \sqrt{4y - y^2} = (4y - y^2)^{1/2}$$

$$\frac{dx}{dy} = \frac{1}{2} (4y - y^2)^{-1/2} \cdot (4 - 2y)$$

$$= \frac{2(2-y)}{2\sqrt{4y-y^2}}$$

$$= \frac{2-y}{\sqrt{4y-y^2}}$$

$$\left(\frac{dx}{dy}\right)^2 = \frac{4-4y+y^2}{4y-y^2}$$

$$1 + \left(\frac{dx}{dy}\right)^2 = \frac{1+4-4y+y^2}{4y-y^2}$$

$$\begin{aligned} 1 + \left(\frac{dx}{dy}\right)^2 &= 1 + \frac{4-4y+y^2}{4y-y^2} \\ &= \frac{4y-y^2+4-4y+y^2}{4y-y^2} \\ &= \frac{4}{4y-y^2} \end{aligned}$$

$$\sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{\frac{4}{4y-y^2}} = \frac{2}{\sqrt{4y-y^2}}$$

$$S = 2\pi \int_1^2 \sqrt{4y-y^2} \cdot \frac{2}{\sqrt{4y-y^2}} dy$$

$$= 4\pi \int_1^2 dy$$

$$= 4\pi [y]_1^2$$

$$= 4\pi (2-1)$$

$$S = 4\pi \text{ sq units}$$